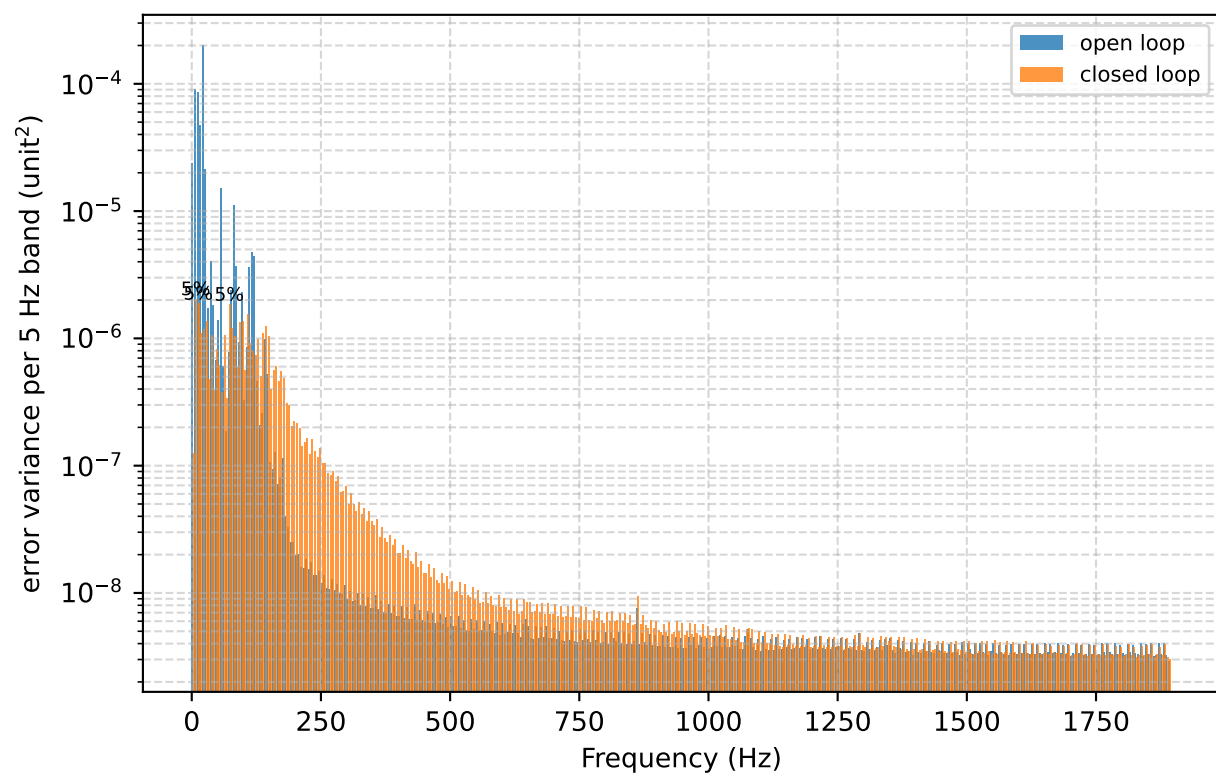
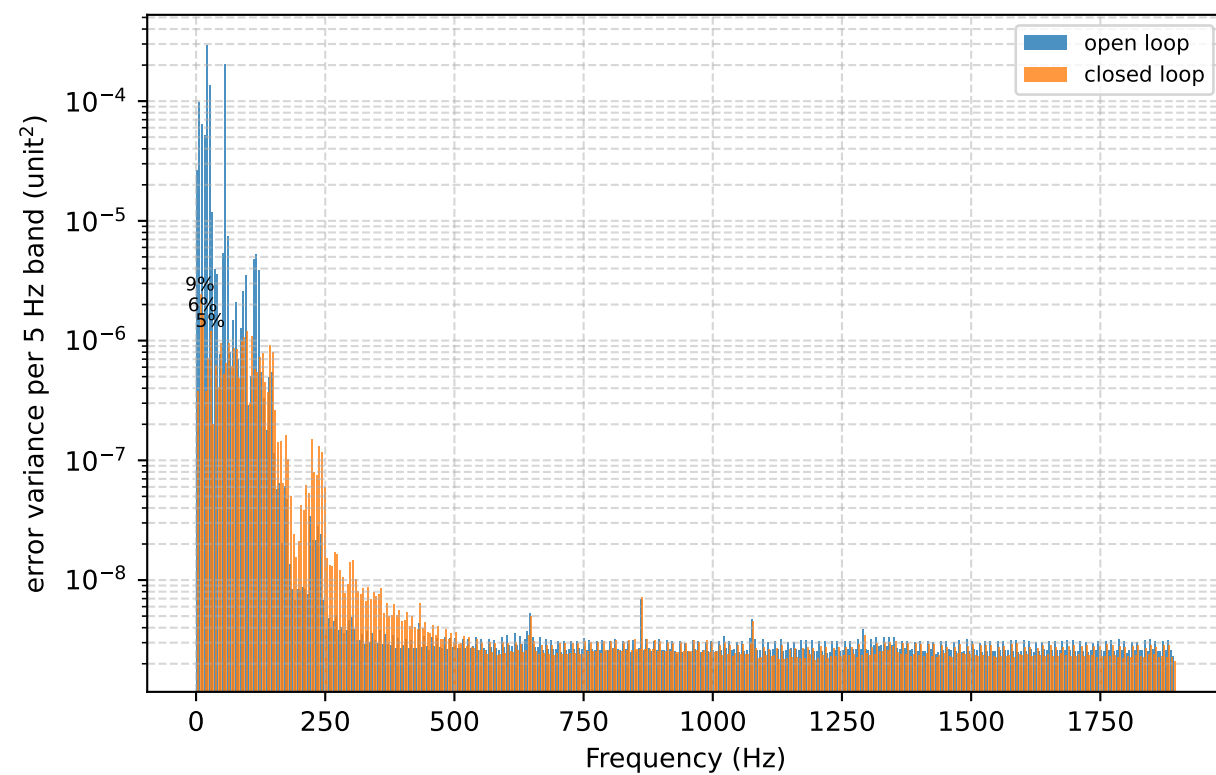
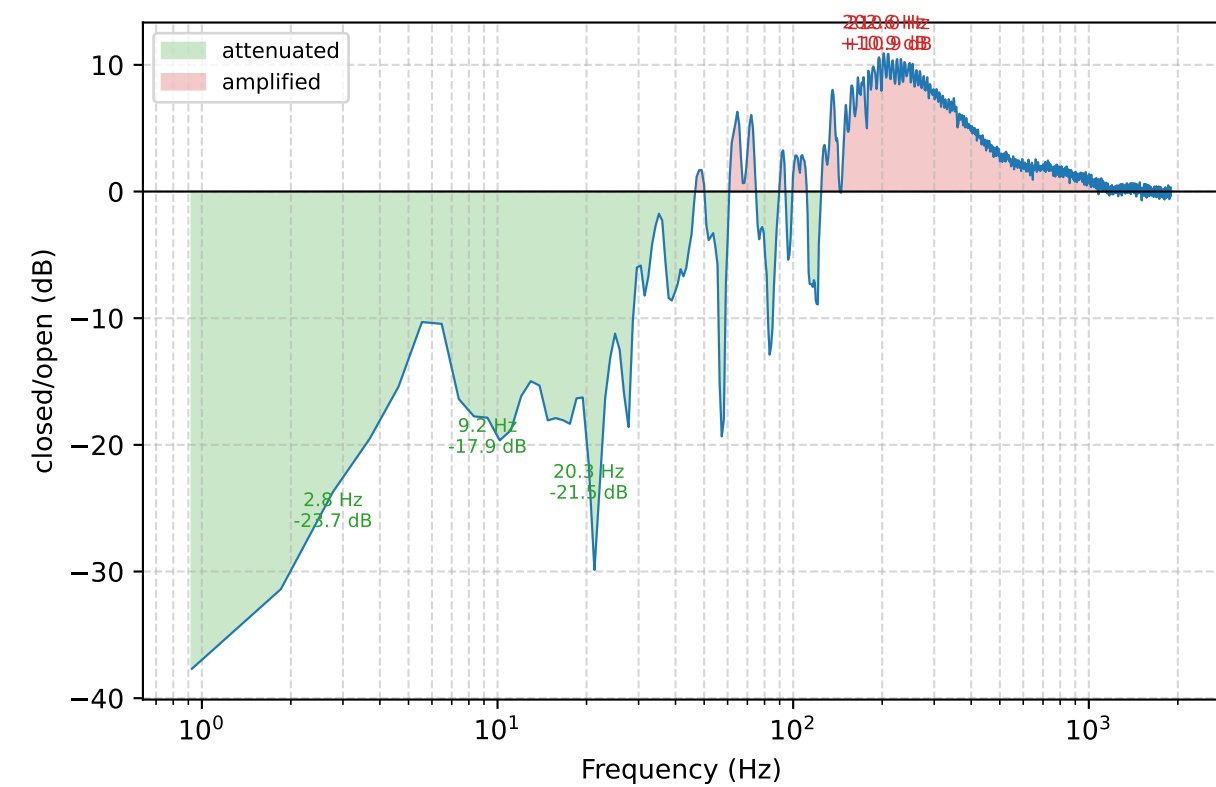
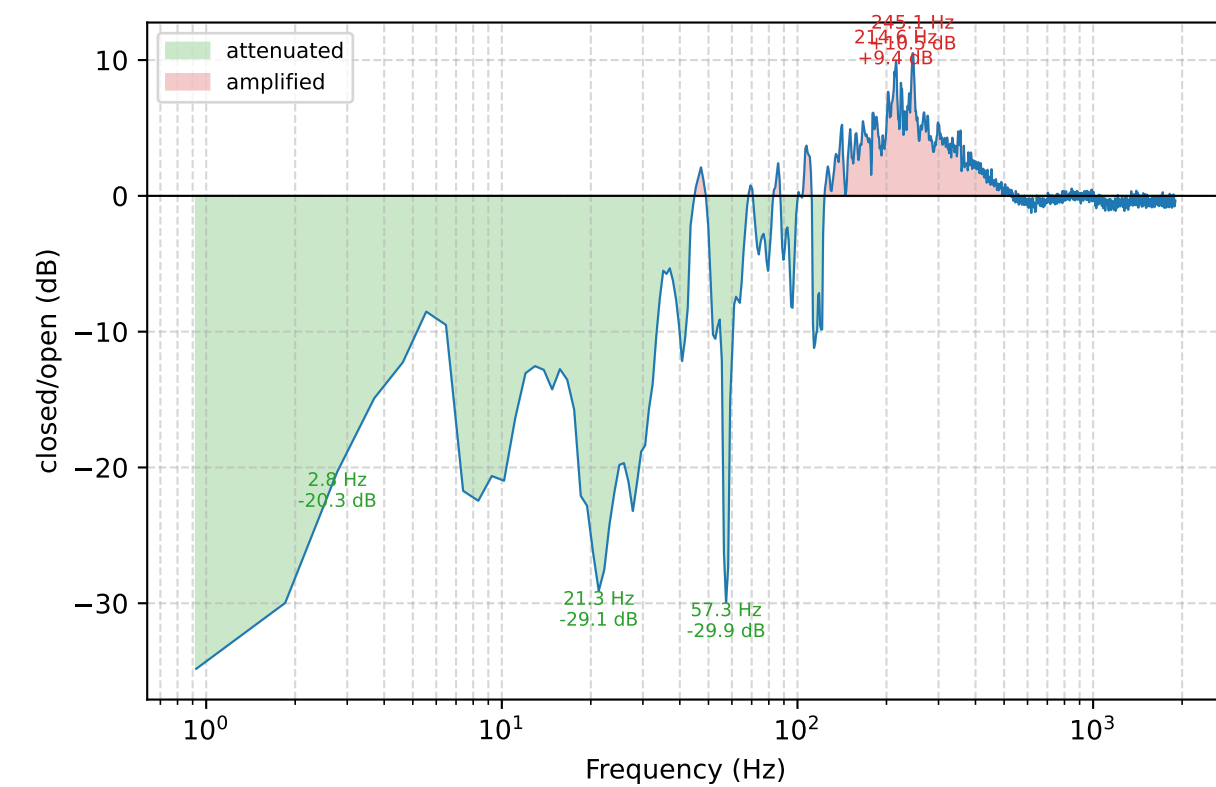
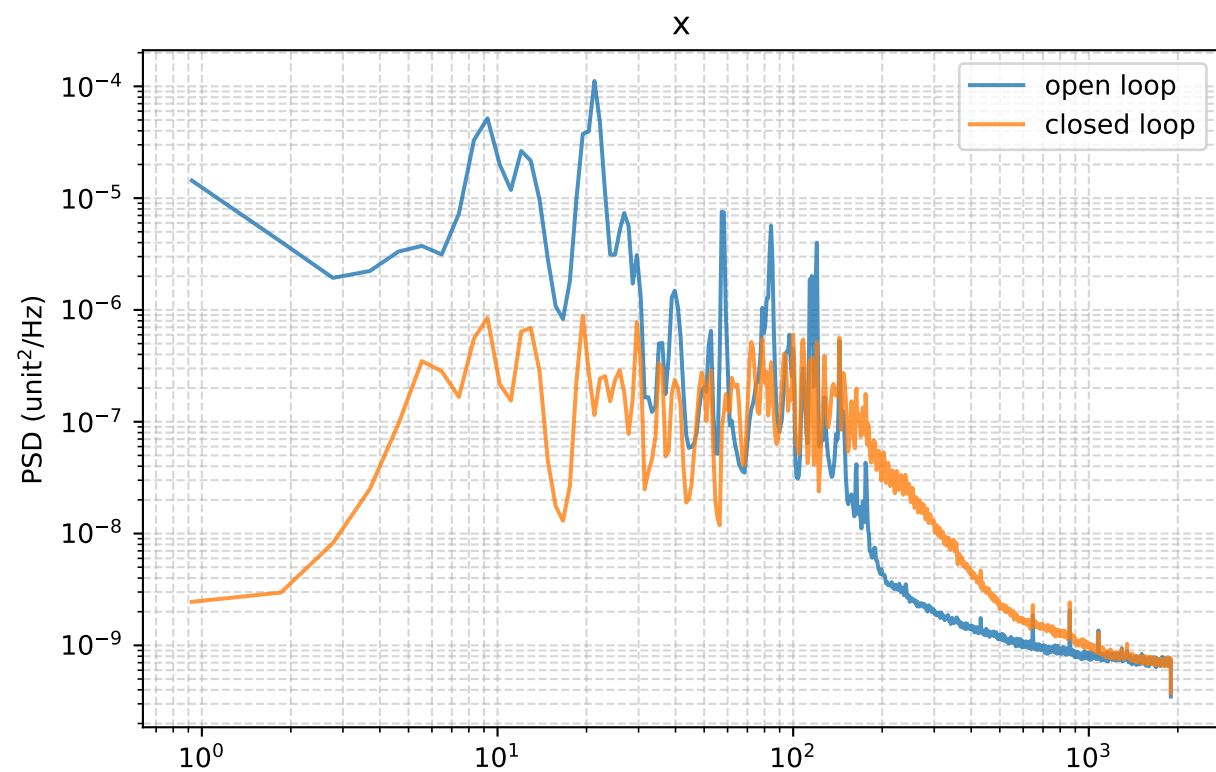
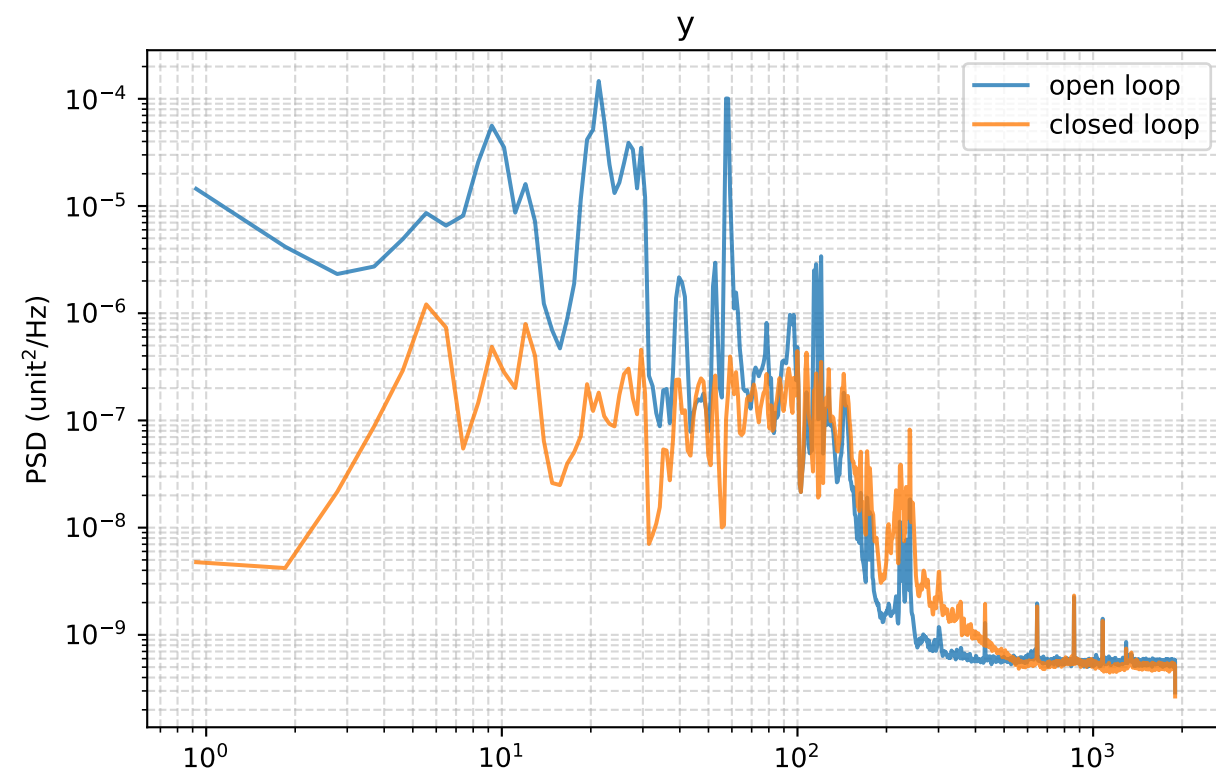


Closed-loop attenuation



Started: 2026-07-30 14:57:12
pixel_scale: 15.5 px per output unit
y: mean +1.748 -> -0.000 px; RMS about mean 0.518 -> 0.080 px (open -> closed)
x: mean +2.540 -> -0.000 px; RMS about mean 0.496 -> 0.097 px (open -> closed)

Config used:
ATTENUATION PAR: 1 -- first closed-loop attenuation run through the PARPORT DAC (TI DAC81404 over the StarTech PEX1P2 / AX99100 PCIe parallel card, spi link, mmio backend), replacing the DAQC2/BRIDGEplate serial path used by every previous attenuation run.
TRANSPORT: bit-timed 24-bit SPI, -6.4 us per channel update (-12.8 us for both channels), ch0 = y/tilt (+1.0 V/unit), ch1 = x/tilt (-1.0 V/unit), offset 0 V, symmetric command bounds +1 V (59 px of authority against a +/-16 px ROI; the mirror's full range is deliberately NOT given to the loop on this first run). NOTE the command->FSM voltage mapping is NOT the same as any previous attenuation run, so K is not comparable across the transport change and was re-measured -- and it came out 4.7-5.5x HIGHER, not lower as this file originally predicted, so nothing about the piplate-era gains carries over.
Camera/CoM: 32x32 ROI start_x 1840 / start_y 680, exposure 50, gain 100, bw 100, 20x20 tracked window thr 5 (the camera is now shrouded, so the background floor no longer needs a high threshold).
FSP: broad_order 512, broad_mu .03, broad_lp 5 (retuned: the shorter delay moved the regeneration band from ~325 Hz to ~700 Hz); adapt_period 8, adapt_tau 5. Guard ON (guard_ratio 4).
DELAY AND K: BOTH re-fit for this transport 2026-07-30 from conf_bode.par x/y.json (+/-100 mV sweep, full-band 5-300 Hz phase slope for tau, <=30 Hz plateau for K). Kx 3.787 +/- 0.008, Ky 3.636 +/- 0.005; delay x 2.717 fr., y 2.577 fr at fs 3788. Against the piplate era that is 4.7-5.5x MORE loop gain and LESS THAN HALF the delay -- the shorter delay is the entire reason to expect better rejection here.
COMPARE against attenuation_fsp21 and attenuation_fsp15 (run 15: y 0.073 / x 0.168 px): report total and per-band closed RMS, 60-400 Hz waterbed, command/clamp behavior, and transient activation count on the shared 500/500/4096 grid. Closed RMS in PIXELS remains comparable across the change; command units do not. refs: Kulcsar 2006 / Petit 2008 / Meimon 2010 / Correia 2010

RMS contributed by each 10 Hz band (px)				
band (Hz)	y open	y closed	x open	x closed
0-10	0.1725	0.0260	0.1666	0.0228
10-20	0.1671	0.0220	0.1787	0.0267
20-30	0.3203	0.0215	0.2303	0.0248
30-40	0.0616	0.0141	0.0372	0.0192
40-50	0.0324	0.0181	0.0245	0.0171
50-60	0.2233	0.0168	0.0631	0.0155
60-70	0.0445	0.0194	0.0138	0.0183
70-80	0.0293	0.0202	0.0276	0.0270
80-90	0.0218	0.0189	0.0599	0.0198
90-100	0.0382	0.0234	0.0280	0.0255
100-110	0.0138	0.0182	0.0168	0.0226
110-120	0.0490	0.0165	0.0450	0.0202
120-130	0.0326	0.0191	0.0342	0.0203
130-140	0.0111	0.0140	0.0106	0.0196
140-150	0.0158	0.0203	0.0190	0.0235
150-160	0.0064	0.0098	0.0070	0.0152
160-170	0.0045	0.0071	0.0069	0.0161
170-180	0.0051	0.0080	0.0068	0.0158
180-190	0.0023	0.0042	0.0042	0.0121
190-200	0.0019	0.0030	0.0035	0.0101
200-210	0.0020	0.0044	0.0031	0.0100
210-220	0.0020	0.0052	0.0028	0.0084
220-230	0.0037	0.0074	0.0029	0.0083
230-240	0.0035	0.0070	0.0027	0.0084
240-250	0.0027	0.0065	0.0026	0.0078
250-260	0.0014	0.0026	0.0025	0.0071
260-270	0.0014	0.0027	0.0023	0.0064
270-280	0.0015	0.0026	0.0024	0.0063
280-290	0.0014	0.0021	0.0023	0.0059
290-300	0.0014	0.0024	0.0023	0.0056
total 0-300	0.4732	0.0785	0.3581	0.0932